

Sentiment Analysis

Text Mining

Text Classification

Remember...

subject domain ?

authorship id ?

language ?

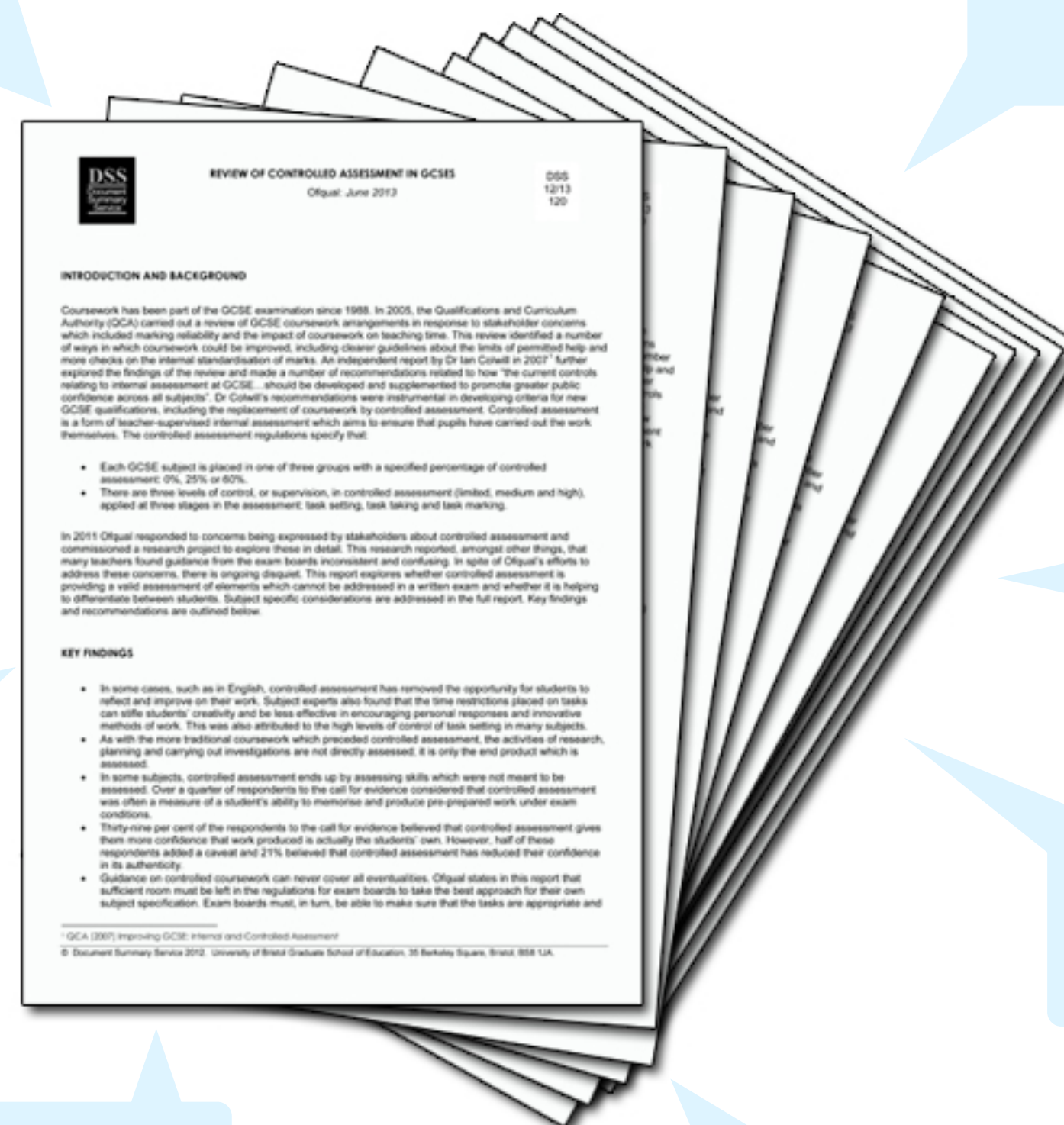
author age ?

sentiment analysis

author gender ?

type of document ?

political orientation?



Text Classification Methods

Remember...

- Hand-coded rules
 - If rules carefully refined by expert, accuracy can be high
 - But building and maintaining these rules may be expensive
- Supervised Machine Learning
 - Assumes a training set of m hand-labeled documents $(d_1, c_1), \dots, (d_m, c_m)$
 - Methods
 - Naïve Bayes
 - Logistic regression
 - Support-vector machines
 - Decision Trees
 - k-Nearest Neighbors
 - ...

Supervised Text Classification: Formal definition

Remember ...

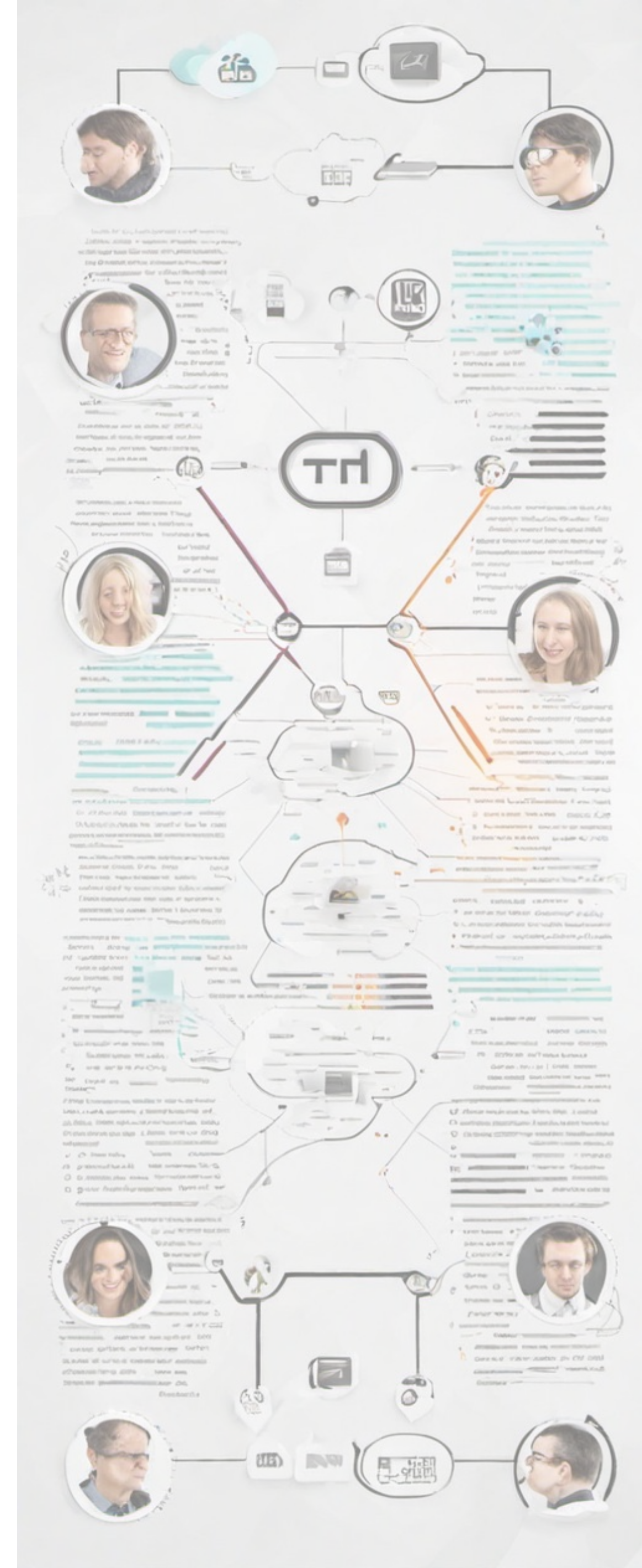
- Text classification is
 - Assigning subject categories, topics, genres, ... to a document
- Input:
 - a document d
 - a fixed set of classes $C = \{c_1, c_2, \dots, c_J\}$
- Output:
 - a learned classifier $\gamma: d \rightarrow c$

Sentiment Analysis

What is Sentiment Analysis?

Application domains

Usage examples



Applications

Sentiment Analysis

- Also known as
 - Opinion extraction
 - Opinion mining
 - Sentiment mining
 - Subjectivity analysis
- Simplest task
 - Is the attitude of this text positive or negative?
- More complex
 - Rank the attitude of this text from 1 to 5
- Advanced
 - Detect the target, source, or complex attitude types

Sentiment Analysis



- Application of Natural Language Processing techniques to identify and extract sentiments and emotions expressed in texts
 - the first step towards the online reputation analysis

Sentiment Analysis



- Application of Natural Language Processing techniques to identify and extract sentiments and emotions expressed in texts
 - the first step towards the online reputation analysis
- Why ?
 - rise of social media, such as blogs and social networks
 - Increasing amount of user-generated contents in the form of reviews, recommendations, ratings and any other form of opinion
 - huge volumes of opinionated texts all over the web
 - whenever we want to make a decision, we want to know what others think



Sentiment Analysis

Application domains

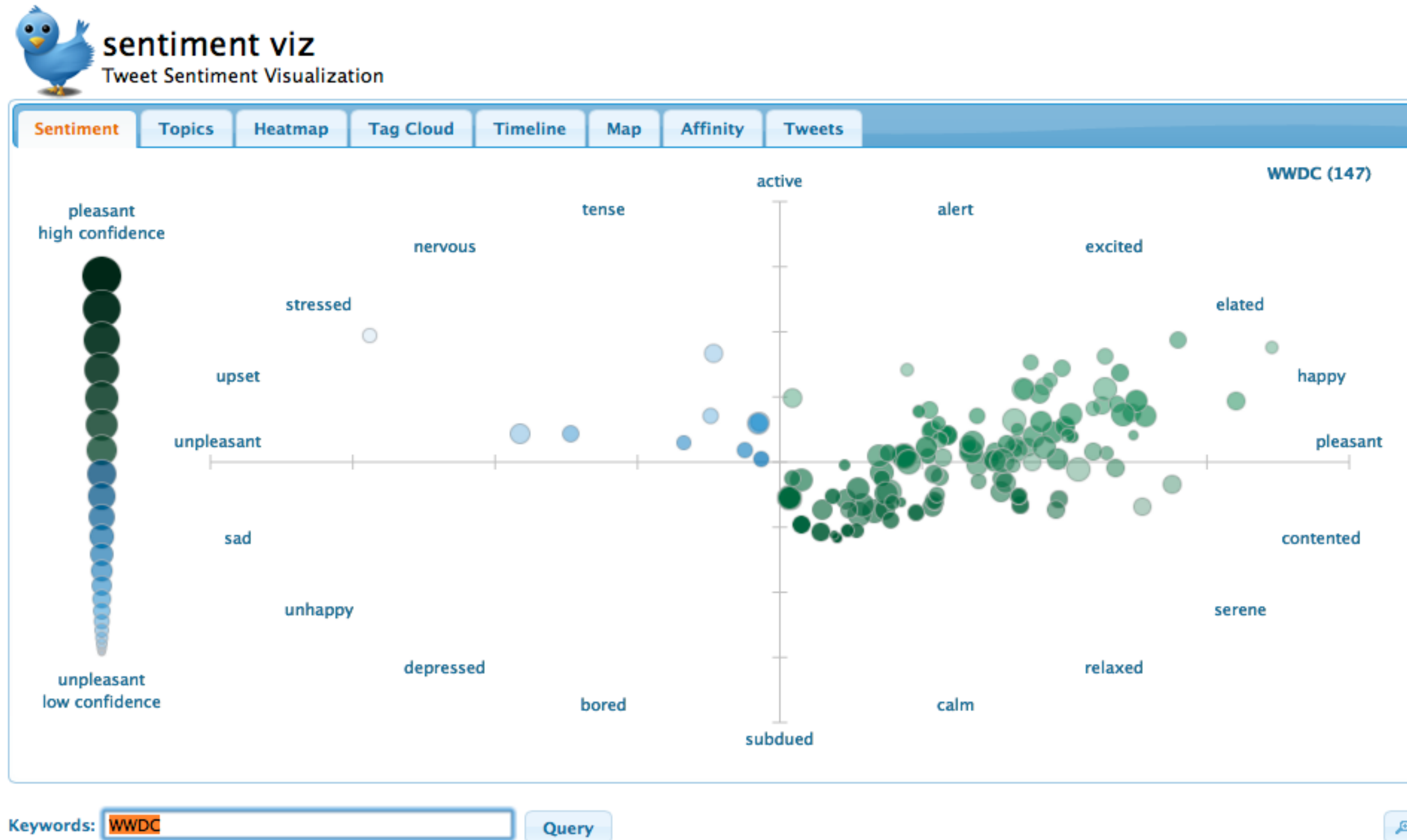
- **Movies**: Is this review positive or negative?
- **Products**: What do people think about the new iPhone?
- **Public sentiment**: How is consumer confidence? Is despair increasing?
- **Politics**: What do people think about this candidate or issue?
- **Prediction**: Predict election outcomes or market trends from sentiment
- **Urban planning**: e.g., Travellers' perspectives (Stellacci, S., & Moro, S., 2023)

Sentiment Analysis Applications



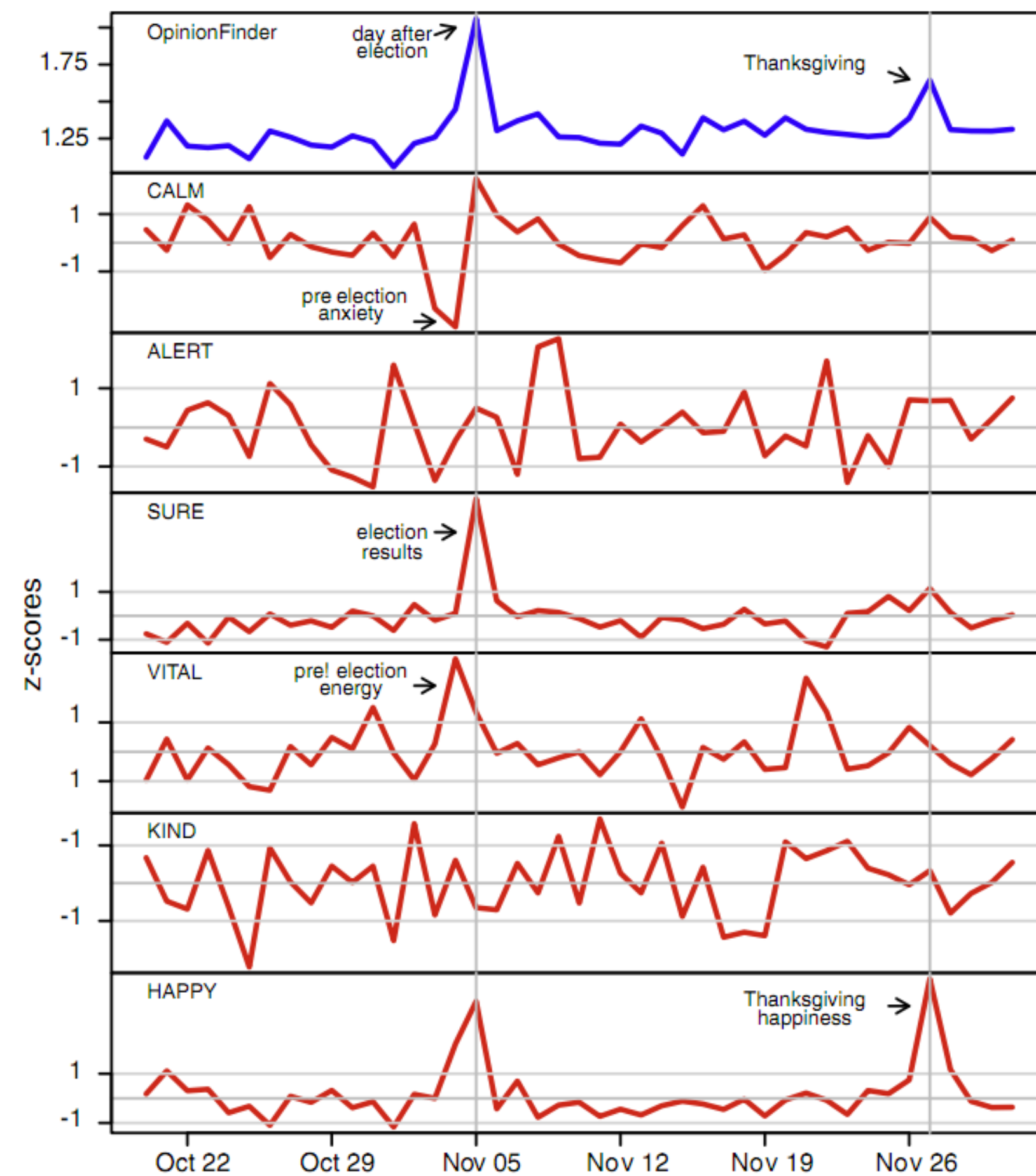
Search performed in 09-06-2013 (THE SITE IS NOW CLOSED)

Sentiment Analysis Applications



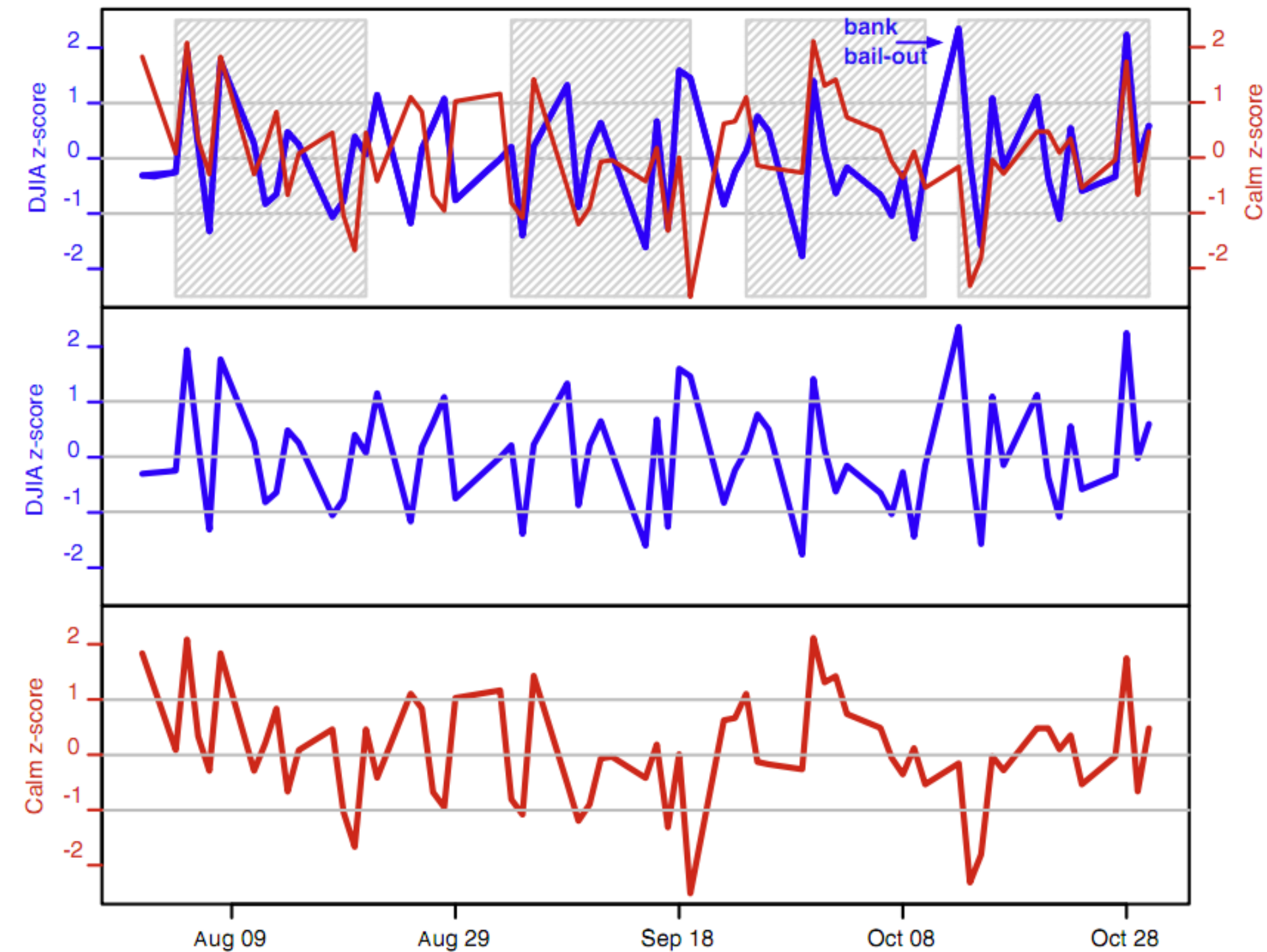
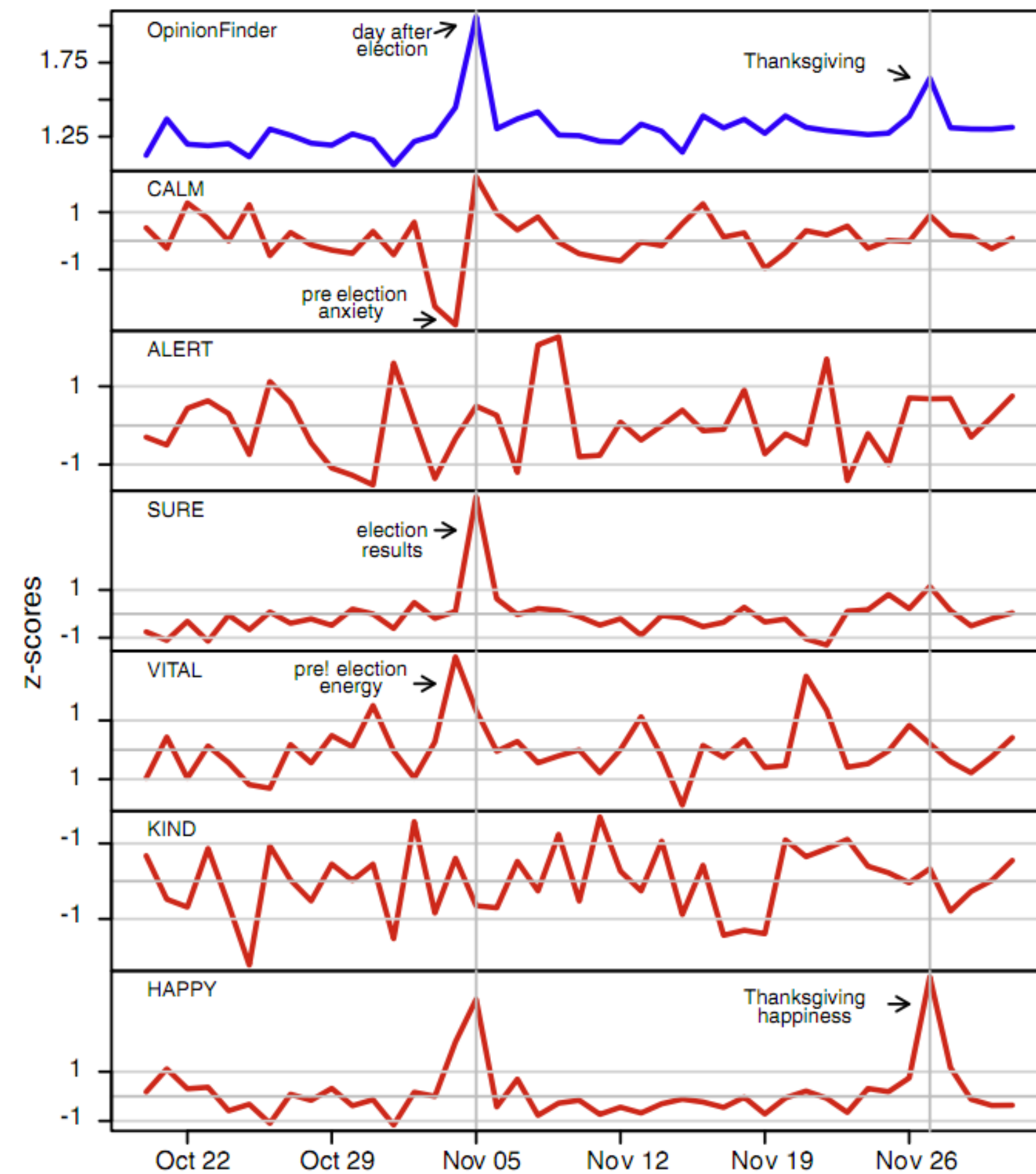
WWDC (Apple Worldwide Developers Conference) – June 10–14, San Francisco – Search performed in June 09 (2013)

Twitter sentiment



Johan Bollen, Huina Mao, Xiaojun Zeng. 2011. Twitter mood predicts the stock market, Journal of Computational Science 2:1, 1-8. 10.1016/j.jocs.2010.12.007.

Twitter sentiment



CALM predicts DJIA 3 days later
At least one hedge fund used this algorithm

Johan Bollen, Huina Mao, Xiaojun Zeng. 2011. Twitter mood predicts the stock market, Journal of Computational Science 2:1, 1-8. 10.1016/j.jocs.2010.12.007.

Grasping emotions and sarcasm...

 **DeepMoji** Artificial emotional intelligence [→ Teach AI!](#) [About](#)

DeepMoji has learned to understand emotions and sarcasm based on millions of emojis. Here's a [video](#) explaining a bit more. Type a sentence to see what our AI algorithm thinks.

I love the plot, but I am sick of it.

SUBMIT

Words are highlighted based on emotional impact. Click a word to turn it on/off.

i love the plot , but i am sick of it .



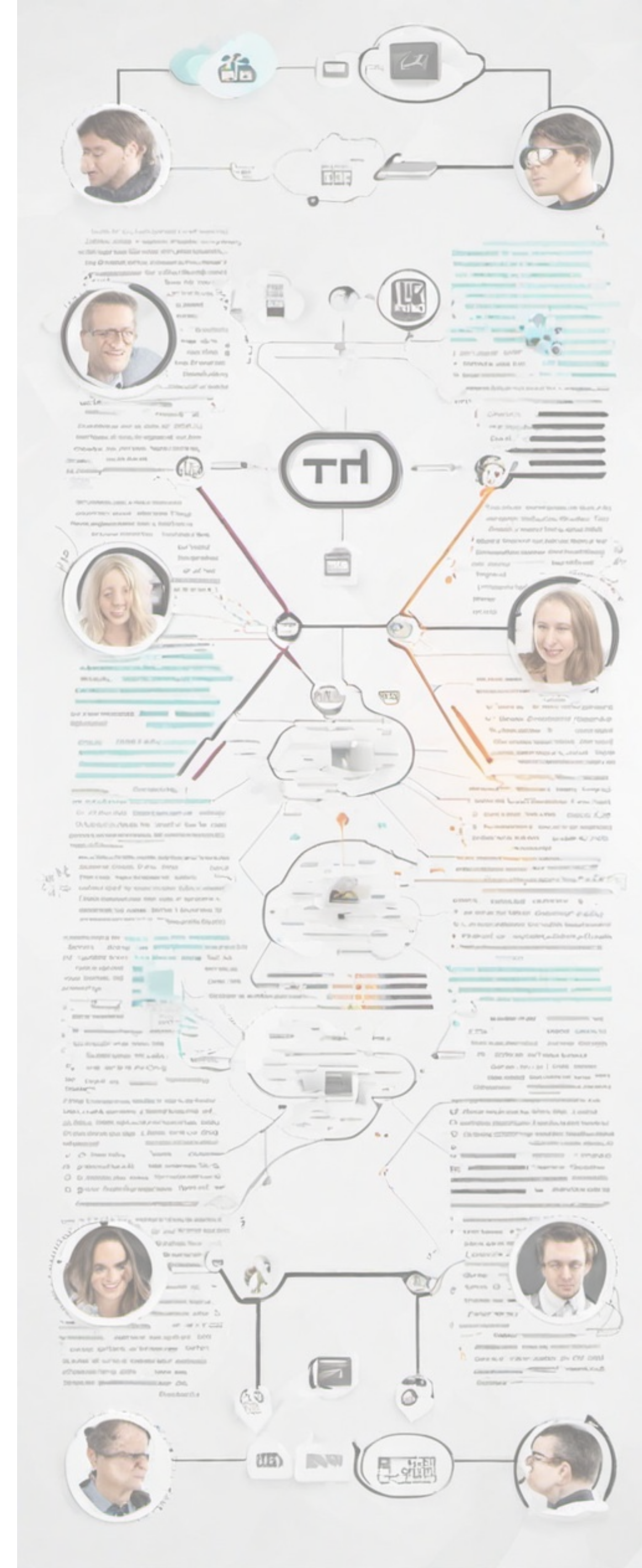
low confidence

Sentiment Analysis

Baseline approach using a sentiment lexicon

Baseline algorithm

Handling negation



Sentiment Classification

Movie Reviews

Bo Pang, Lillian Lee, and Shivakumar Vaithyanathan. 2002. Thumbs up? Sentiment Classification using Machine Learning Techniques. EMNLP-2002, 79—86.
Bo Pang and Lillian Lee. 2004. A Sentimental Education: Sentiment Analysis Using Subjectivity Summarization Based on Minimum Cuts. ACL, 271-278

- Polarity detection:
 - Is an IMDB movie review positive or negative?
- Data: Polarity Data 2.0:
 - <http://www.cs.cornell.edu/people/pabo/movie-review-data>

Sentiment Analysis

IMDB data in the Pang and Lee database

when _star wars_ came out some twenty years ago , the image of traveling throughout the stars has become a commonplace image . [...]

when han solo goes light speed , the stars change to bright lines , going towards the viewer in lines that converge at an invisible point .

cool .

october sky offers a much simpler image—that of a single white dot , traveling horizontally across the night sky . [. . .]

“ snake eyes ” is the most aggravating kind of movie : the kind that shows so much potential then becomes unbelievably disappointing .

it’s not just because this is a brian depalma film , and since he’s a great director and one who’s films are always greeted with at least some fanfare .

and it’s not even because this was a film starring nicolas cage and since he gives a brauvara performance , this film is hardly worth his talents .

Sentiment Analysis

Baseline algorithm (adapted from Pang and Lee)

1. Tokenization & pre-processing
2. Feature Extraction
3. Classification using different classifiers
 - Naïve Bayes
 - Logistic regression
 - Support-vector machines

Sentiment Analysis

Tokenization and related issues

- Deal with HTML and XML markup
- Twitter mark-up (names, hash tags)
- Capitalization (preserve for words in all caps)
- Phone numbers, dates
- Emoticons

Sentiment Analysis

Feature extraction

- Which words to use?
 - Only adjectives (convey emotion)?
 - All words?
 - All words turns out to work better sometimes

Sentiment Analysis

Handling negation

- How to handle negation?

I **didn't** like this movie

vs

I really like this movie

Sentiment Analysis

Handling negation

- How to handle negation?

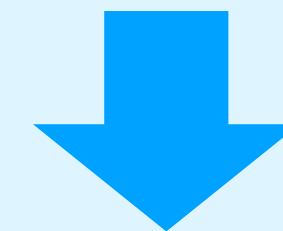
I **didn't** like this movie

vs

I really like this movie

Add **NOT_** to every word between negation and the following punctuation

I didn't like this movie, but I



I didn't NOT_like NOT_this NOT_movie but I

Sanjiv Das and Mike Chen (2001). Yahoo! for Amazon: Extracting market sentiment from stock message boards. In Proceedings of the Asia Pacific Finance Association Annual Conference (APFA).

Bo Pang, Lillian Lee, and Shivakumar Vaithyanathan (2002). Thumbs up? Sentiment Classification using Machine Learning Techniques. In Proceedings of EMNLP'2002, 79—86.

Sentiment Analysis

Sentiment Lexicons

- The General Inquirer (<http://www.wjh.harvard.edu/~inquirer>)
 - Positive (1915 words) and Negative (2291 words)
 - Strong vs Weak, Active vs Passive, Overstated versus Understated
 - Pleasure, Pain, Virtue, Vice, Motivation, Cognitive Orientation, etc
 - Free for Research Use
- LIWC - Linguistic Inquiry and Word Count (<http://www.liwc.net/>)
 - 2300 words, >70 classes
 - Affective Processes
 - negative emotion (bad, weird, hate, problem, tough); positive emotion (love, nice, sweet)
 - Cognitive Processes
 - Tentative (maybe, perhaps, guess), Inhibition (block, constraint)
 - Pronouns, Negation (no, never), Quantifiers (few, many)

Sentiment Analysis

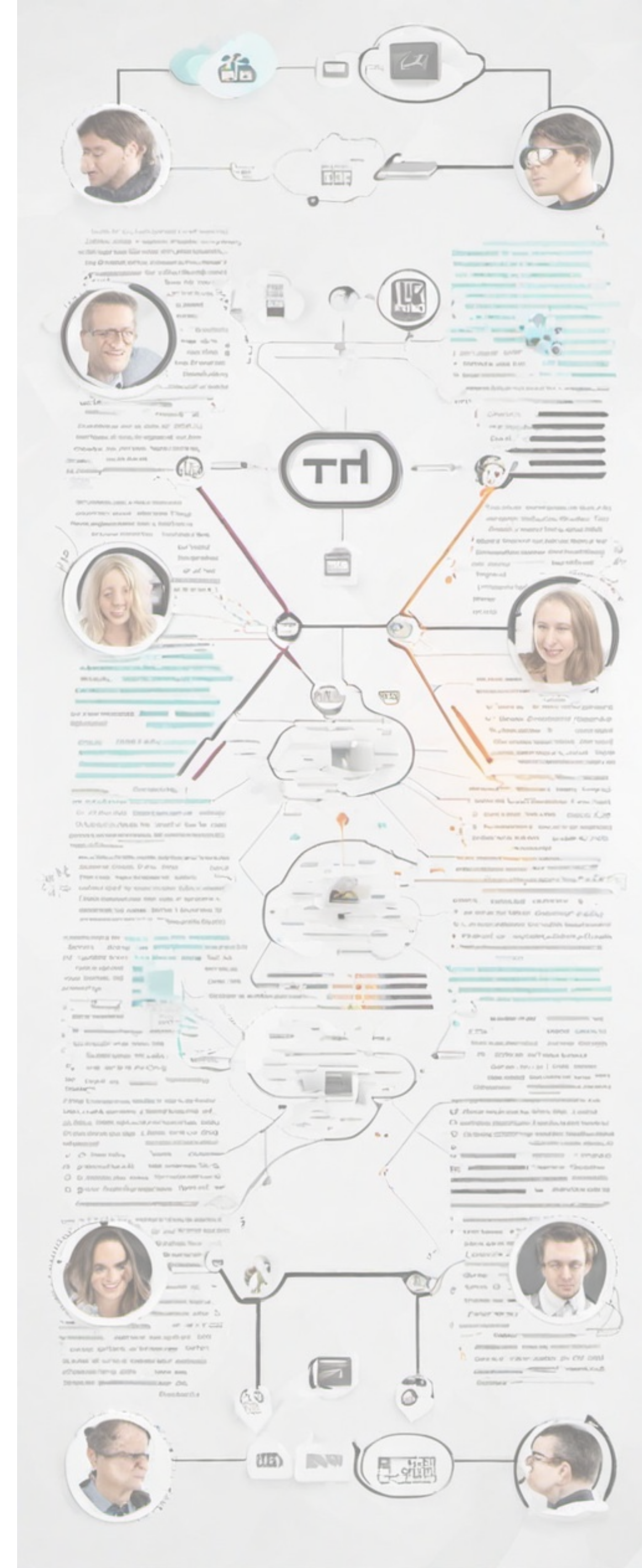
More challenges: what makes reviews hard to classify?

- Subtlety
 - Perfume review in Perfumes: the Guide
 - *If you are reading this because it is your **darling fragrance**, please **wear it at home exclusively, and tape the windows shut**.*
- Thwarted expectations and ordering effects
 - Movie reviews
 - *This film should be **brilliant**. It sounds like a **great** plot, the actors are **first grade**, and the supporting cast is good as well, and Stallone is attempting to deliver a **good** performance. **However, it can't hold up**.*
 - *Well as usual Keanu Reeves is nothing special, but surprisingly, the **very talented** Laurence Fishbourne is **not so good** either, I was surprised.*

Sentiment Analysis

Supervised Learning

Case study



Sentiment Analysis

Supervised Text Classification

- Text => Features => Text Classification (Machine Learning)

Case Study

- TASS 2012 (2013, 2014) - Workshop on Sentiment Analysis
 - Promote the research of **sentiment analysis** and **topic classification** in **social media**
 - Promote the application of existing and new state-of-the-art algorithms and techniques on **short text opinions** extracted from social media messages, specifically Twitter
 - specially focused on Spanish language

Case Study

Data

Training data	Test data
7 200 tweets (labeled)	61 000 tweets (unlabelled)

- Tweets written in Spanish
 - By 200 well-known personalities and celebrities of the world of politics, economy, communication, mass media and culture
 - Between November 2011 and March 2012.
- XML with information about the users, including the user type, which assumes three possible values
 - Periodista (journalist), famoso (famous person), and politico (politician)

Case Study

Data annotations

- Each tweet was manually annotated with 6 possible polarity levels
 - **N, N+** negative sentiment (+ means its more intense)
 - **P, P+** positive sentiment (+ means its more intense)
 - **NEU** (neutral) contains both positive and negative sentiments
 - **NONE** without sentiment information
- Each tweet was annotated with one or more topics, from a list of 10 possible topics
 - política (politics), entretenimiento (entertainment), economía (economics), música (music), fútbol (football), cine (movies), tecnología (technology), deportes (sports), literatura (literature), otros (others)

Case Study

Strategies and methods

- Most relevant NLP tasks
 - Spelling error correction
 - Lemmatisation
 - POS tagging
 - Use of named entities and adjectives
 - Syntactic information
 - Word sense disambiguation
 - Emoticons
 - Processing of negation
- Polarity lexicons
- Twitter-LDA - Latent Dirichelet Allocation (for Topic Classification)
- Information retrieval-motivated (based on language divergence)
- Classification methods
 - Logistic Regression
 - Multinomial Naive Bayes
 - SVM (Support Vector Machines)

Case Study

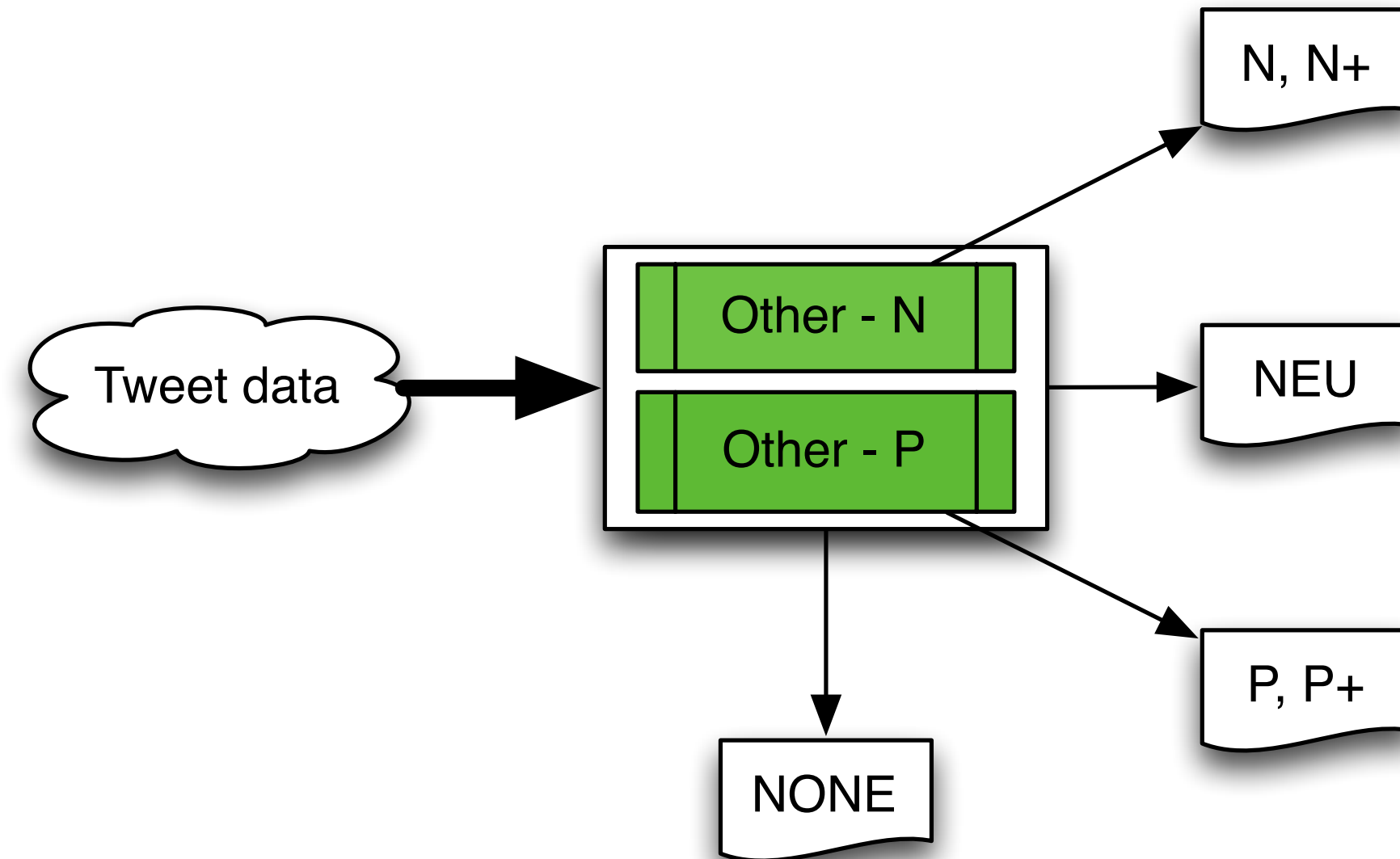
Our approach

- Both tasks (sentiment analysis and topic classification) using the same approach (**classification tasks**)
 - Maximum Entropy Models
 - Provide probability distributions over classes
 - Combine **binary** classifications (discriminate between two classes)
- Sentiment Lexicons in Spanish (Perez-Rosas et al., 2012).
 - Only the most robust part was used
 - `fullStrengthLexicon`, containing 1346 words, automatically labelled with sentiment polarity

Case Study

Architecture for Sentiment Analysis

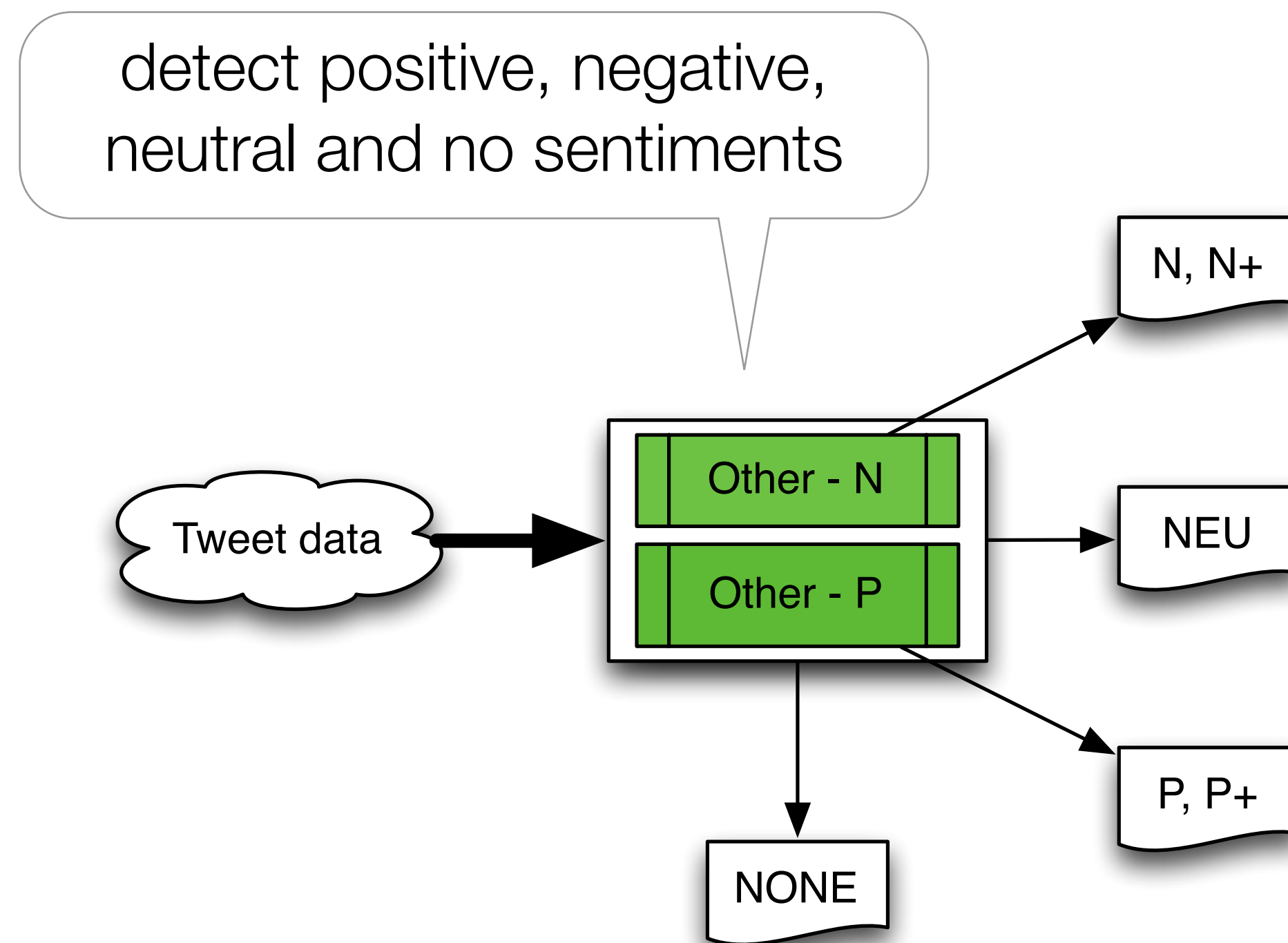
- Our results were achieved by combining 4 different binary classifiers



Case Study

Architecture for Sentiment Analysis

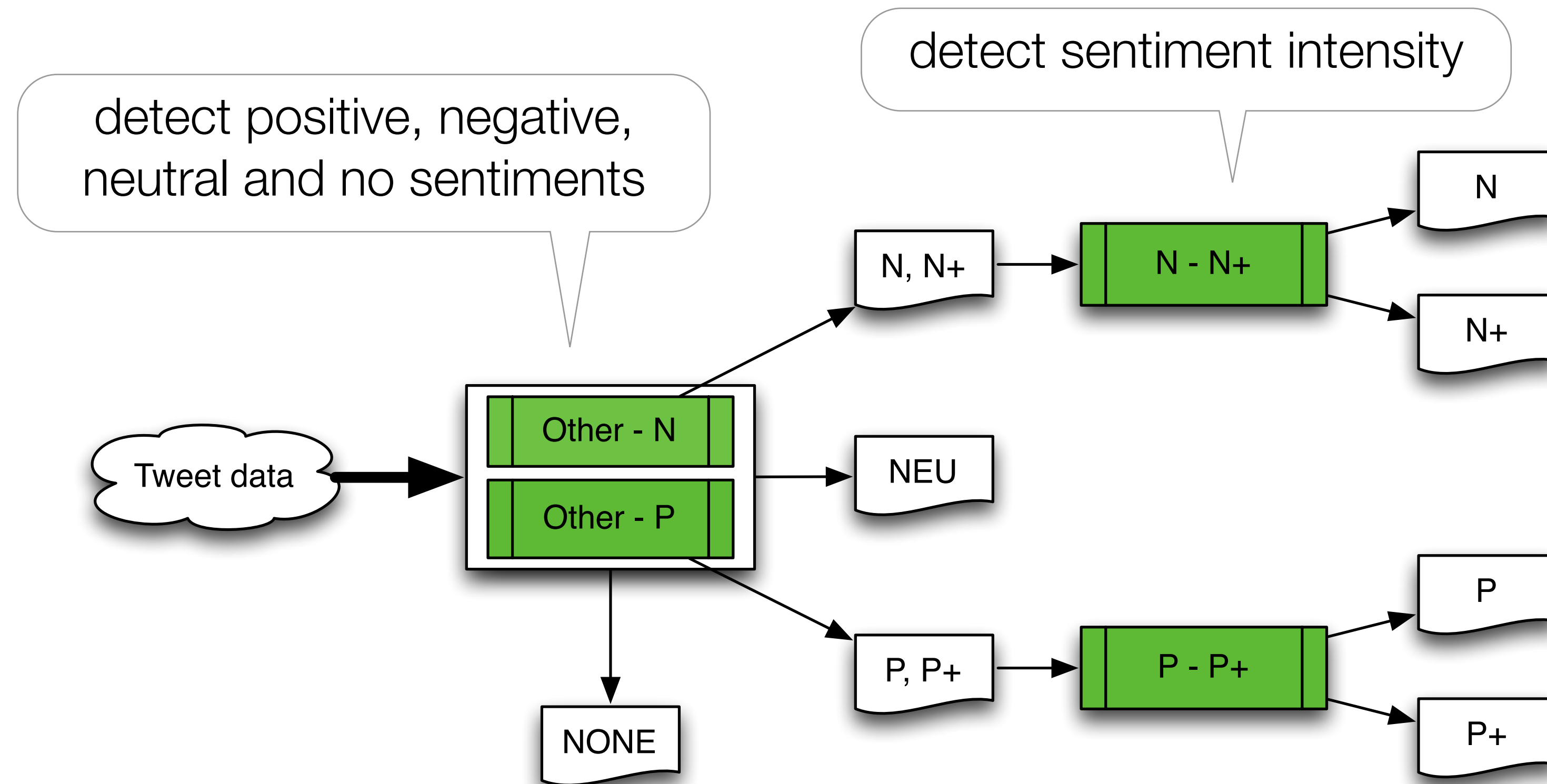
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Case Study

Architecture for Sentiment Analysis

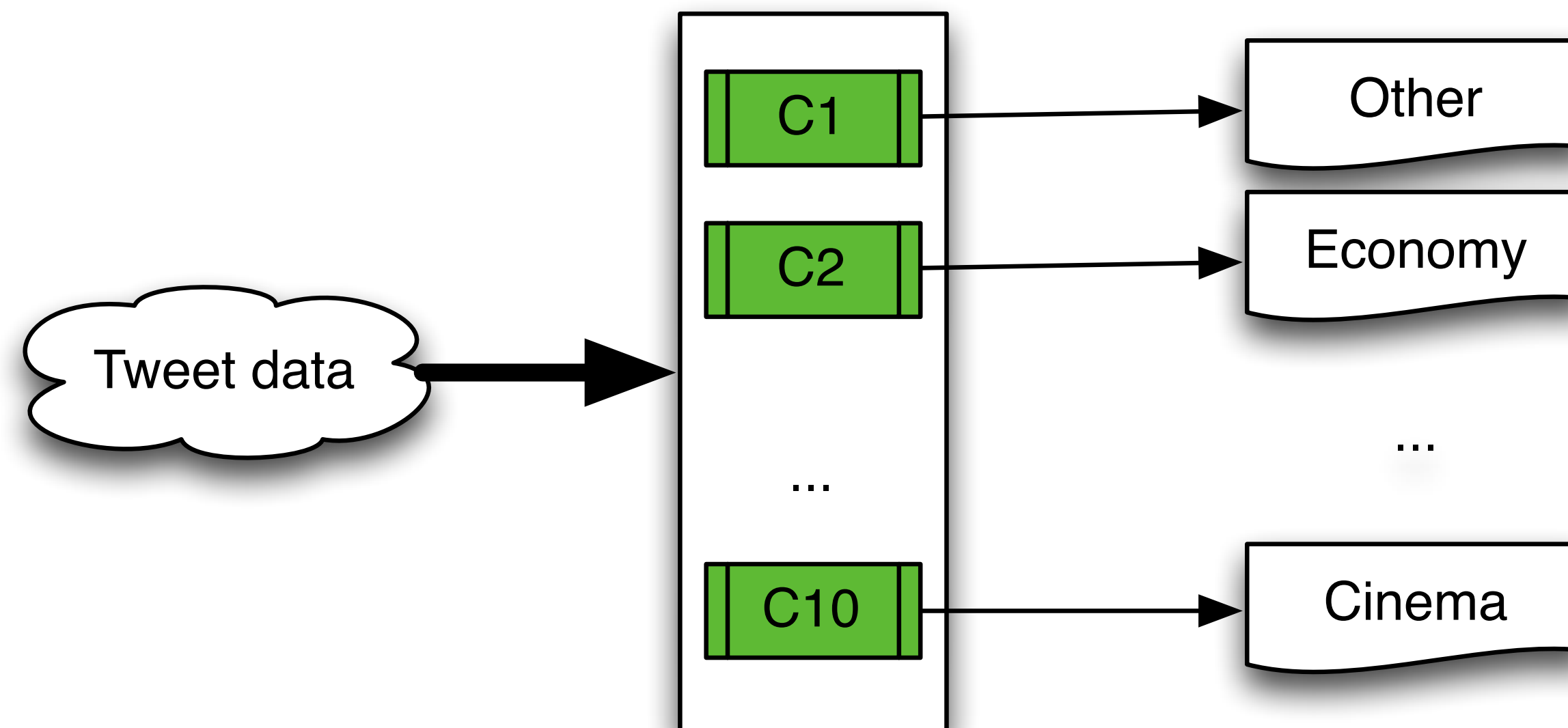
- Our results were achieved by combining 4 different binary classifiers



Case Study

Architecture for Topic Classification

- 10 distinct binary classifiers, each one for a different topic
- Each classifier selects its corresponding topic, and in the case no topic was selected, the most probable topic is then selected based on the available classification probabilities



Case Study

Selected text features

- Punctuation marks
- All words after the words "nunca" (never) or "no" (no) prefixed by **NO_** until reaching some punctuation mark or until reaching the end of the tweet
- Hashtags used as plain text
- All words with more than 3 repeating letters, generating two features
 - **LONG_WORD**, and the corresponding word without repetitions

It was a **loooooong** day! → It was a **long LONG_WORD** day!

- All cased words, and the corresponding lowercase words
 - Uppercase words were assigned a higher weight, since they are often used for emphasis

Case Study

Results

- Sentiment analysis

- Best Approach

- Treated emoticons, negation, spelling errors, lemmatisation, POS-tagging, syntactic analysis, sentiment lexicons

- Our approach

- Text features + tweet's author name (+1.1%) + user type (+0.6%)

Best results (accuracy)

Sentiment analysis		Topic Classification
5 levels	3 levels	
65,3%	71,1%	65,4%
63.4%	69.1%	60,1%
57%	62%	45,3%
...		...

Case Study

Results

- Sentiment analysis

- Best Approach

- Treated emoticons, negation, spelling errors, lemmatisation, POS-tagging, syntactic analysis, sentiment lexicons

- Our approach

- Text features + tweet's author name (+1.1%) + user type (+0.6%)

- Topic Classification

- Our approach

- Text features (without punctuation marks) + tweet's author name
 - Topic-based evaluation: 78% accuracy

Best results (accuracy)

Sentiment analysis		Topic Classification
5 levels	3 levels	
65,3%	71,1%	65.4%
63.4%	69.1%	60,1%
57%	62%	45,3%
...		...

Case Study

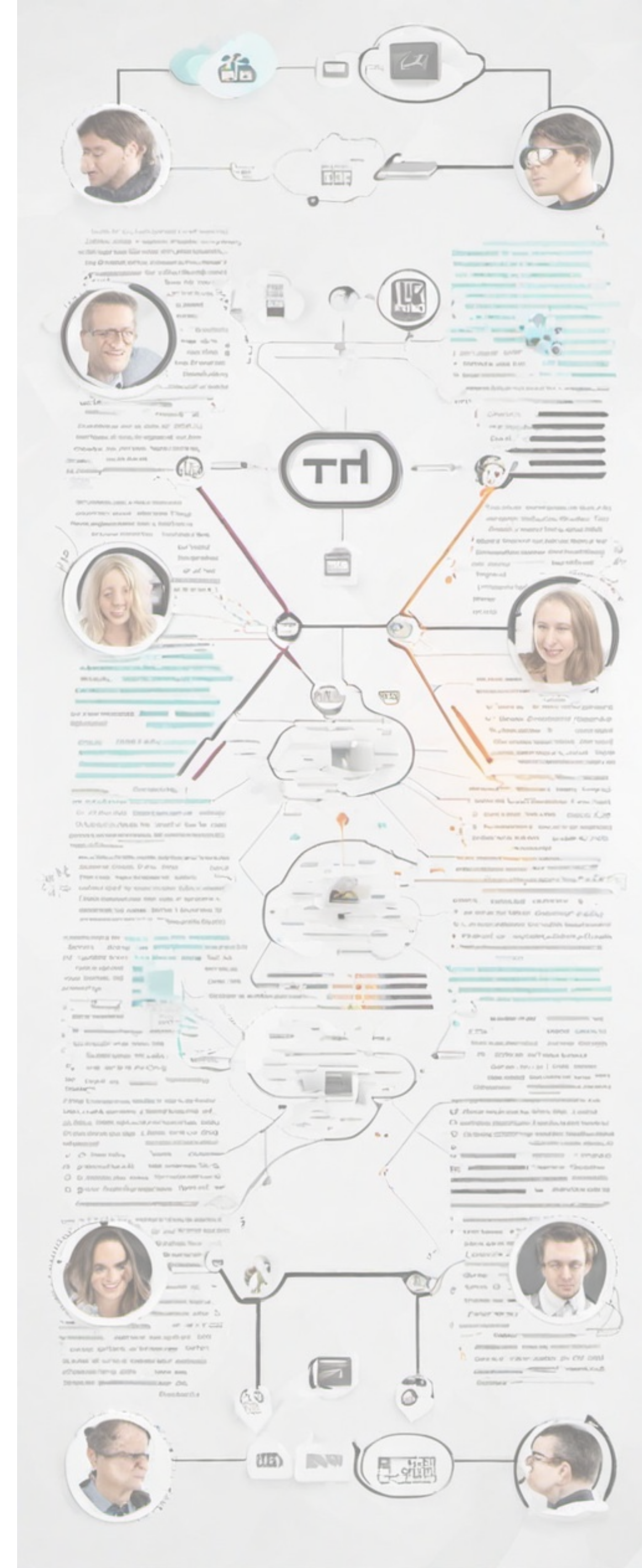
Summary and conclusions

- Case study on Twitter data
 - Sentiment analysis possible at Twitter document level (tweet)
 - Twitter messages are small and therefore tend to focus on a particular issue
 - Is 65% accuracy (71% for four levels) an acceptable performance?
 - Topic Classification is relatively easy to perform (78% accuracy)
 - Depends on a pre-defined set of topics

Sentiment Analysis

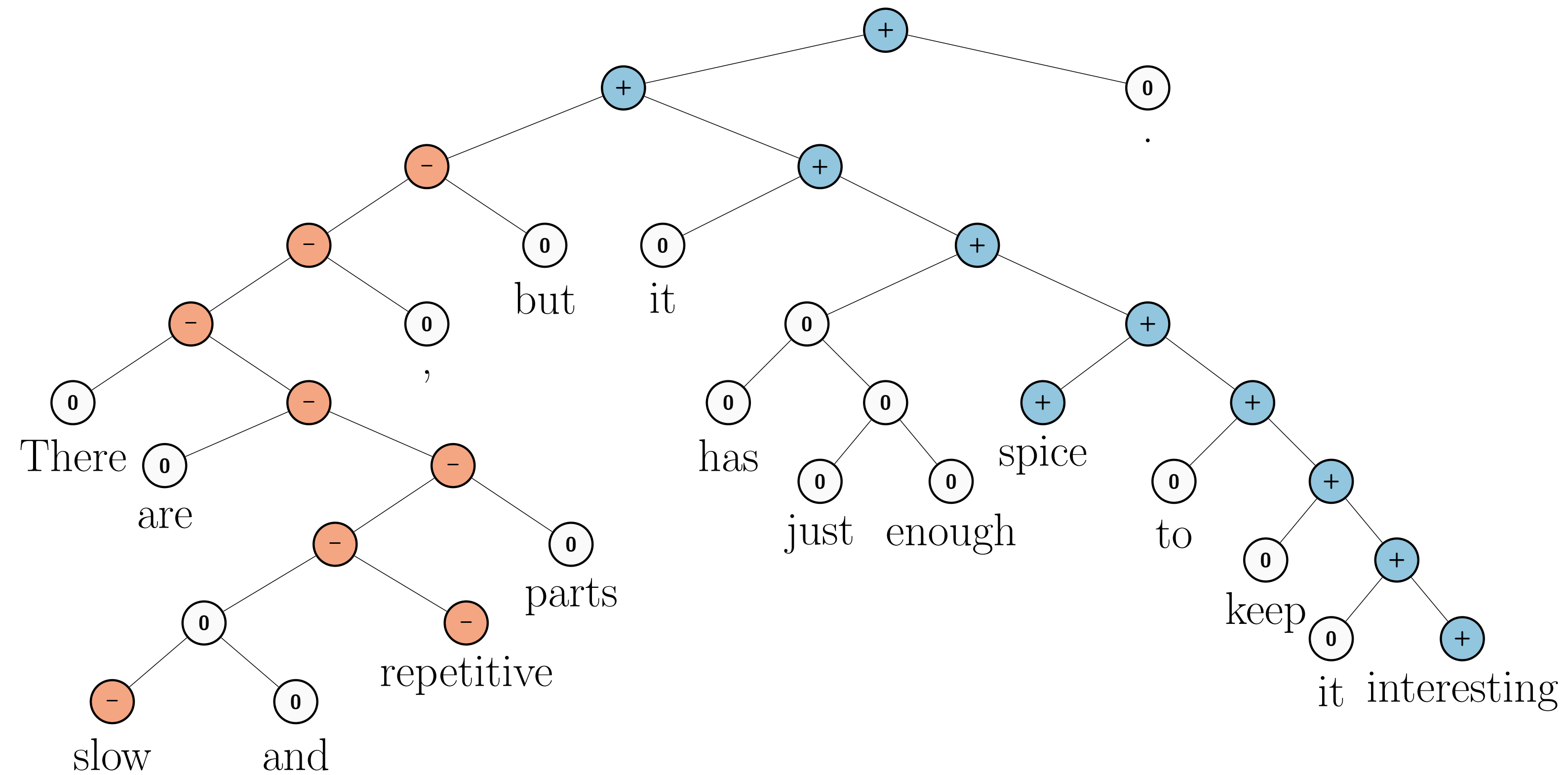
Transformers and Large Language Models

Case study



Sentiment Analysis

Deep Learning



Socher et. Al (2013) Recursive Deep Models for Semantic Compositionality Over a Sentiment Treebank

Transformers ...

Attention Is All You Need

Ashish Vaswani*

Google Brain

avaswani@google.com

Noam Shazeer*

Google Brain

noam@google.com

Niki Parmar*

Google Research

nikip@google.com

Jakob Uszkoreit*

Google Research

usz@google.com

Llion Jones*

Google Research

llion@google.com

Aidan N. Gomez* †

University of Toronto

aidan@cs.toronto.edu

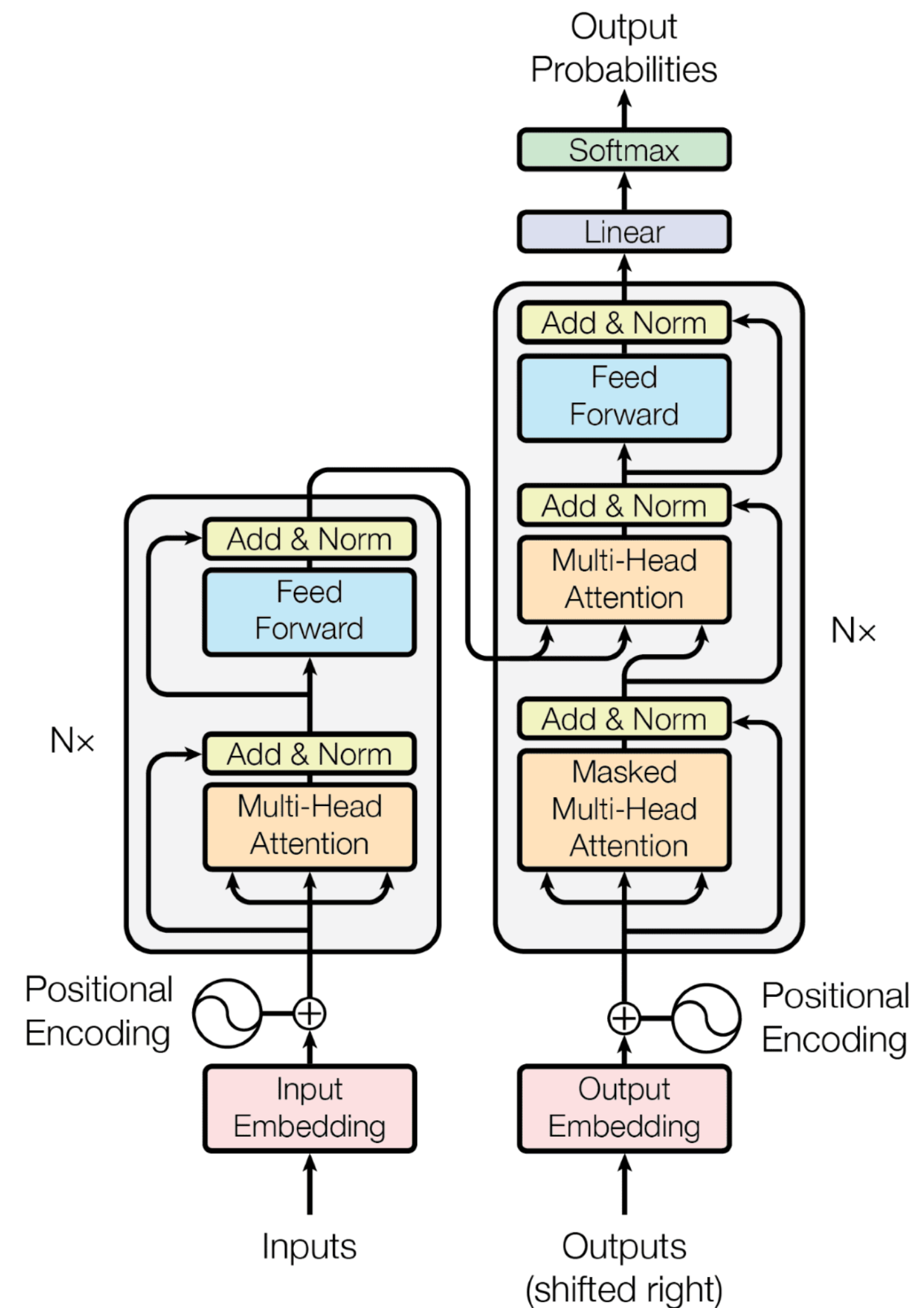
Łukasz Kaiser*

Google Brain

lukaszkaizer@google.com

Illia Polosukhin* ‡

illia.polosukhin@gmail.com



Transformers ...

Attention Is All You Need

Ashish Vaswani*
Google Brain
avaswani@google.com

Noam Shazeer*
Google Brain
noam@google.com

Niki Parmar*
Google Research
nikip@google.com

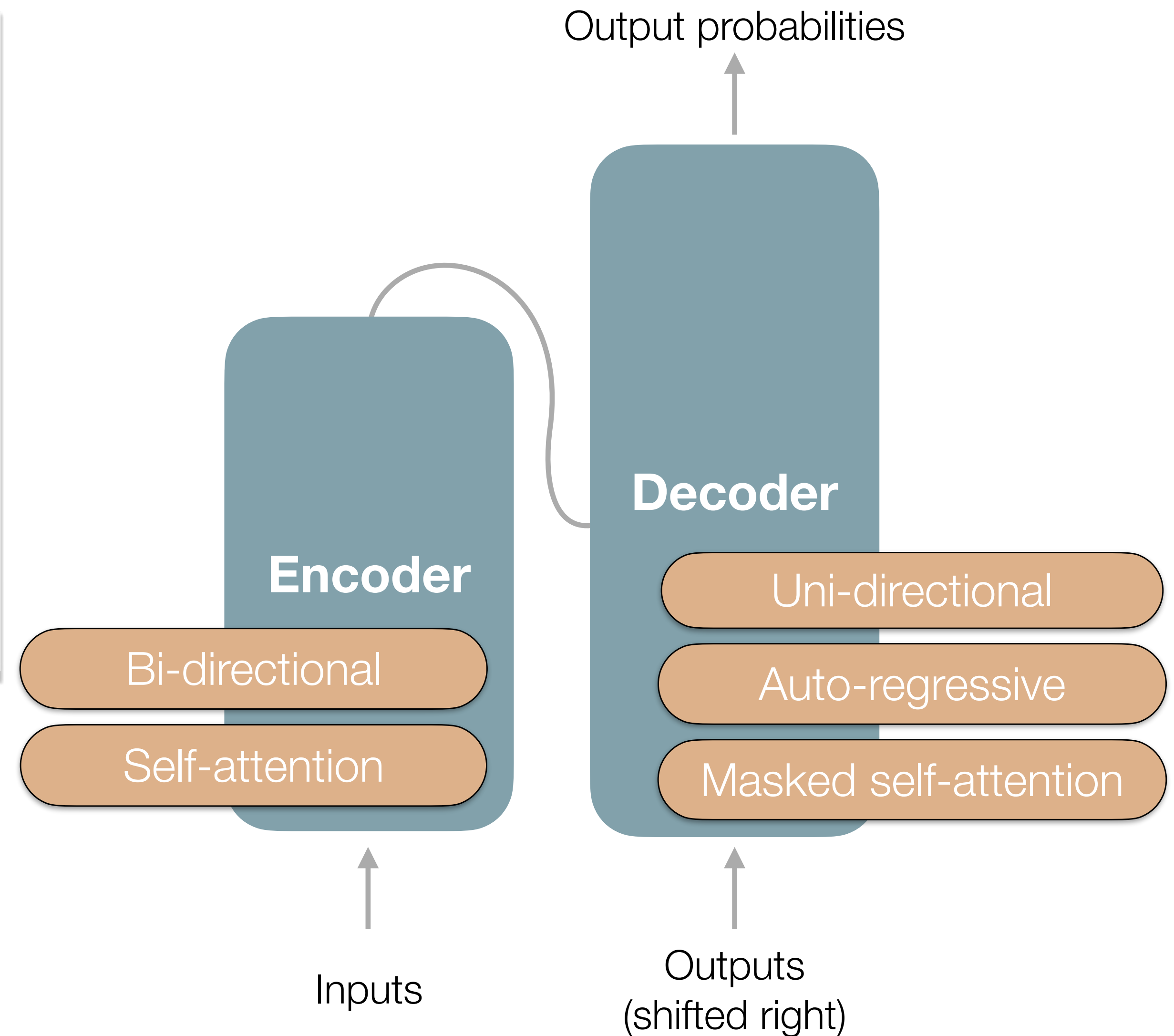
Jakob Uszkoreit*
Google Research
usz@google.com

Llion Jones*
Google Research
llion@google.com

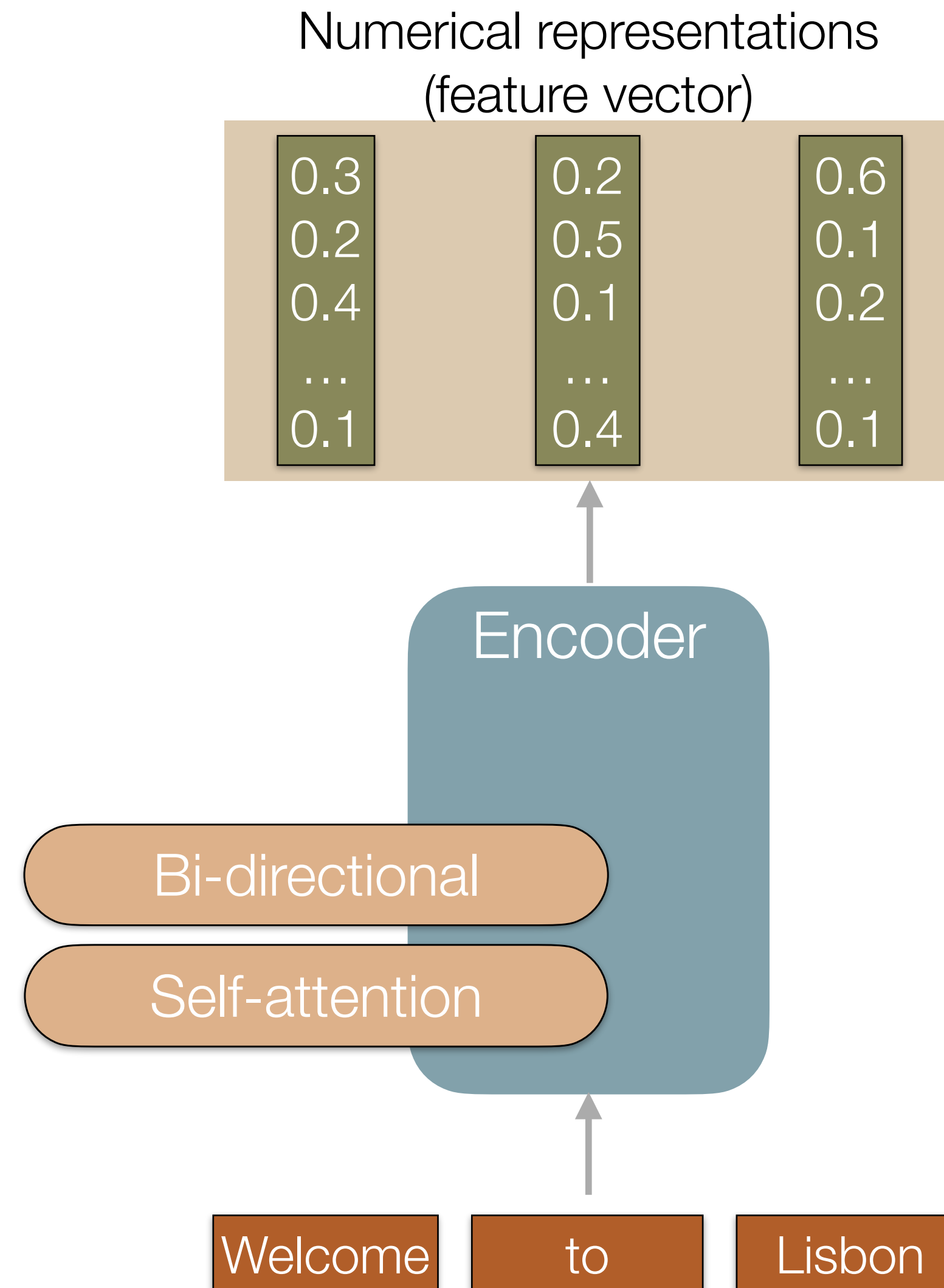
Aidan N. Gomez* †
University of Toronto
aidan@cs.toronto.edu

Łukasz Kaiser*
Google Brain
lukaszkaiser@google.com

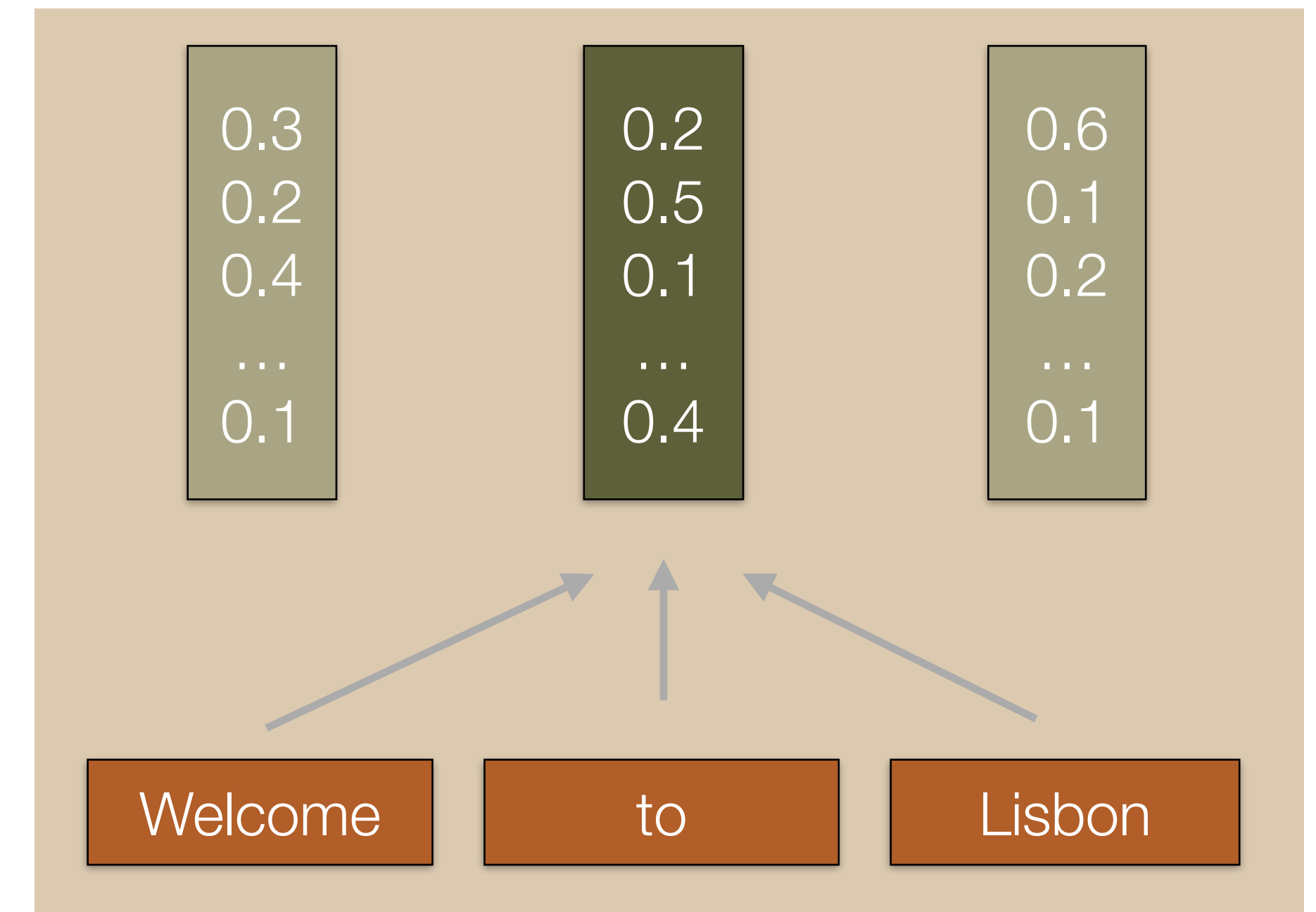
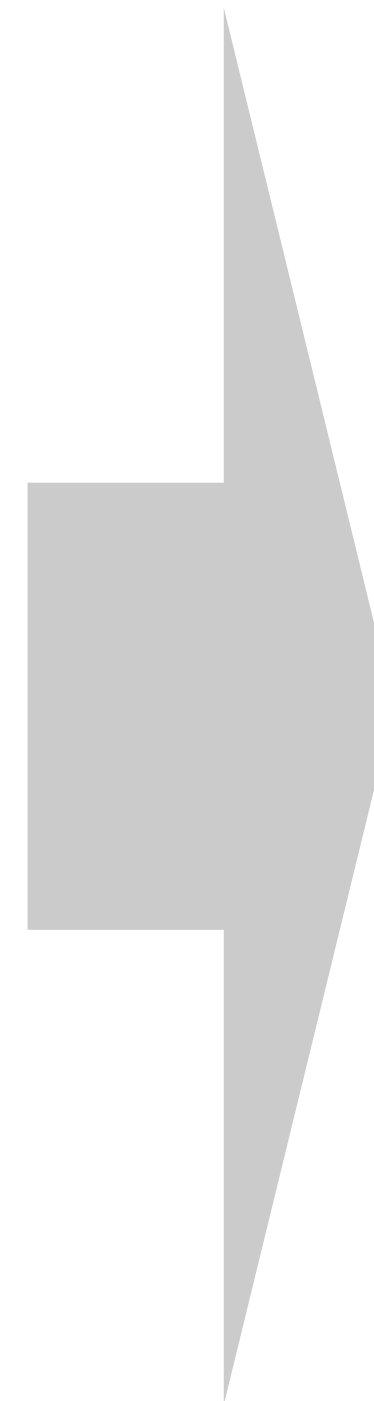
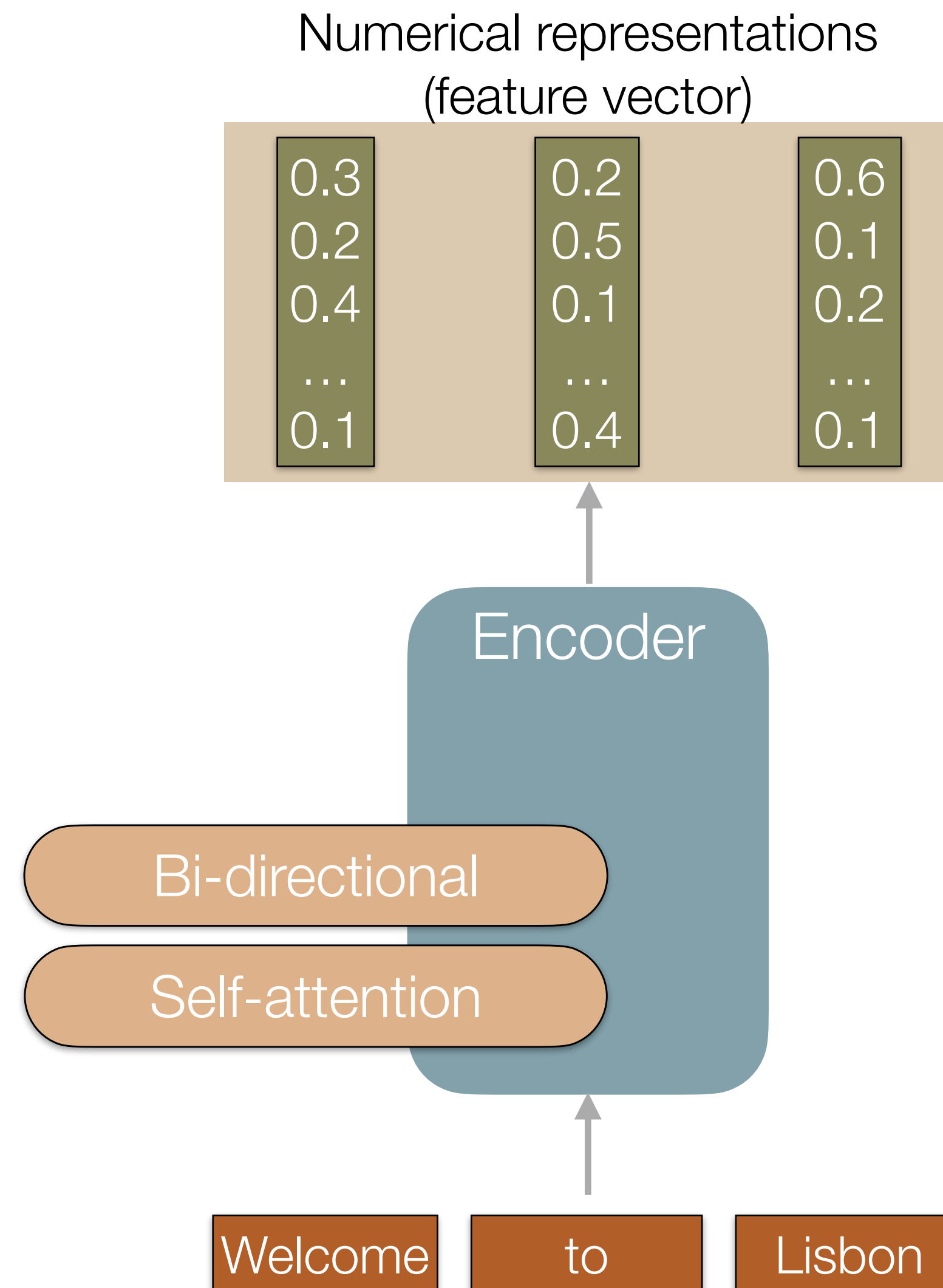
Illia Polosukhin* ‡
illia.polosukhin@gmail.com



Transformer architecture: Encoder



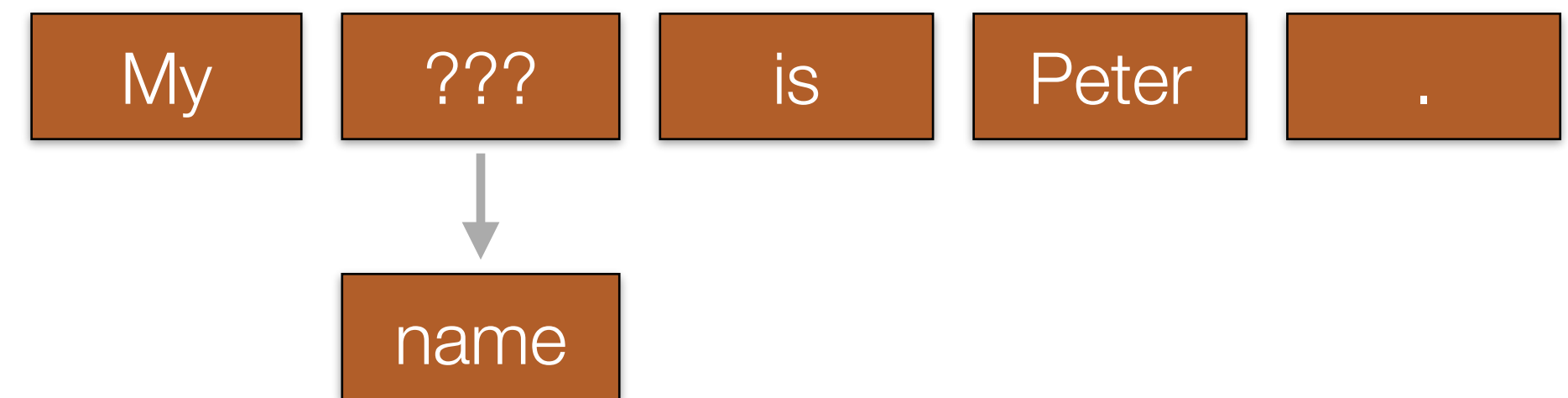
Transformer architecture: Encoder



Contextualised representation: Every word in the initial sequence affects every word's representation

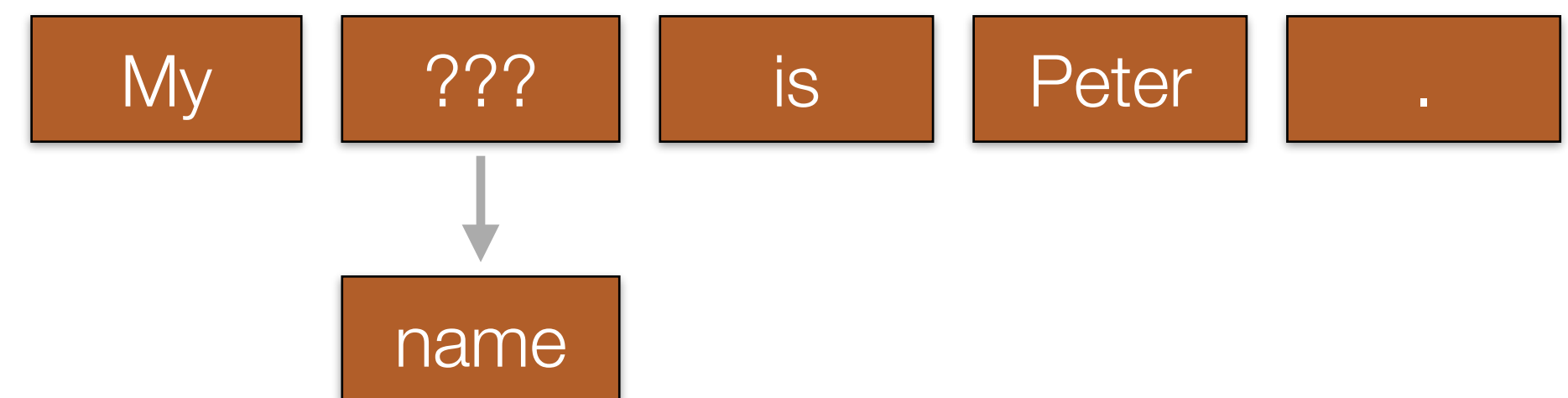
Examples where Encoders are really good

- Filling the blanks
 - Encoders, with their bi-directional context, are good for **guessing the words in the middle of a sequence**



Examples where Encoders are really good

- Filling the blanks
 - Encoders, with their bi-directional context, are good for **guessing the words in the middle of a sequence**



- Sequence classification (e.g. Sentiment Analysis)
 - Encoders are good at obtaining an **understanding of sequences**

Even though I am sad to see them go, I am grateful. **positive**

I am sad to see them go, I can't be grateful. **negative**

Recordar...

O que é possível fazer com um *transformador* “*Encoder*”?

- Um modelo pré-treinado pode ser facilmente adaptado para realizar várias tarefas

Recordar...

O que é possível fazer com um *transformador* “*Encoder*”?

- Um modelo pré-treinado pode ser facilmente adaptado para realizar várias tarefas

- Sentiment Analysis

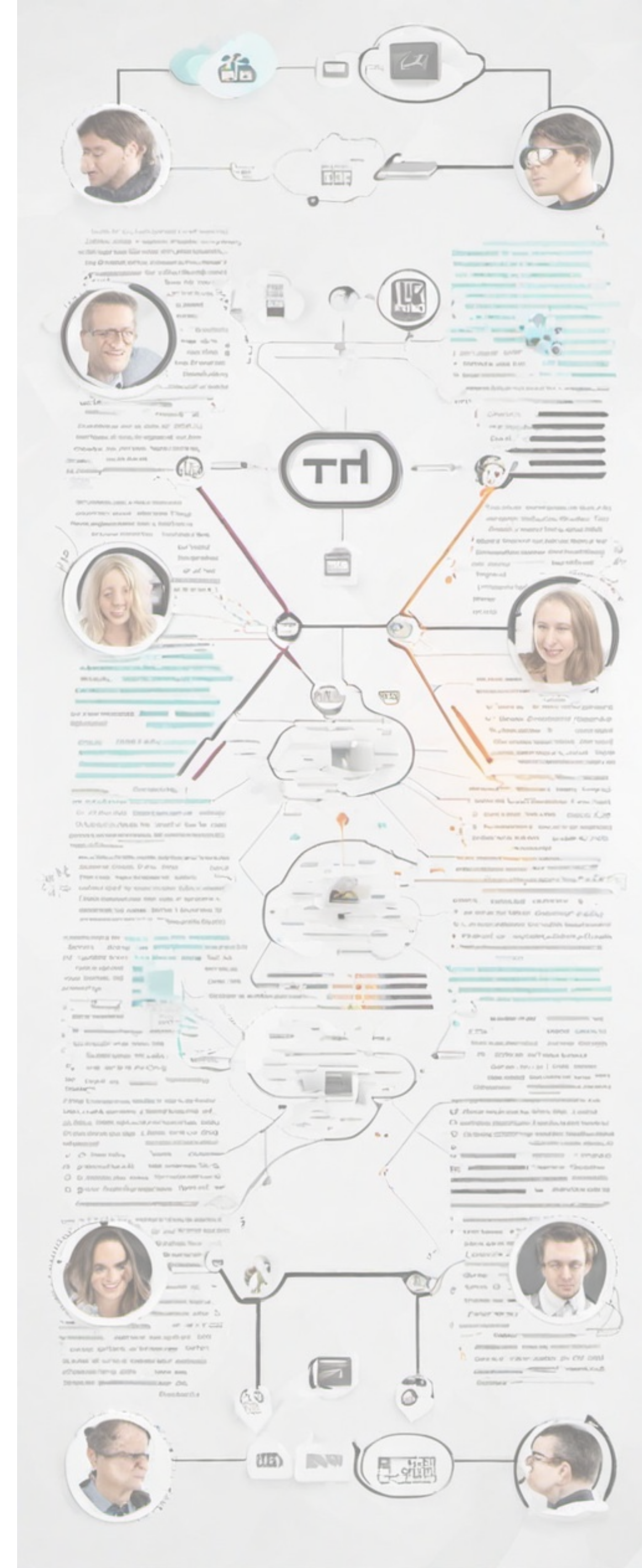
```
from transformers import pipeline
classifier = pipeline("sentiment-analysis")
classifier(["I've been waiting for this opportunity my whole life.", "I hate my cat."])

[{'label': 'POSITIVE', 'score': 0.9969557523727417},
 {'label': 'NEGATIVE', 'score': 0.9994049072265625}]
```


Inteligência Artificial Generativa

Impacto no desenvolvimento de aplicações

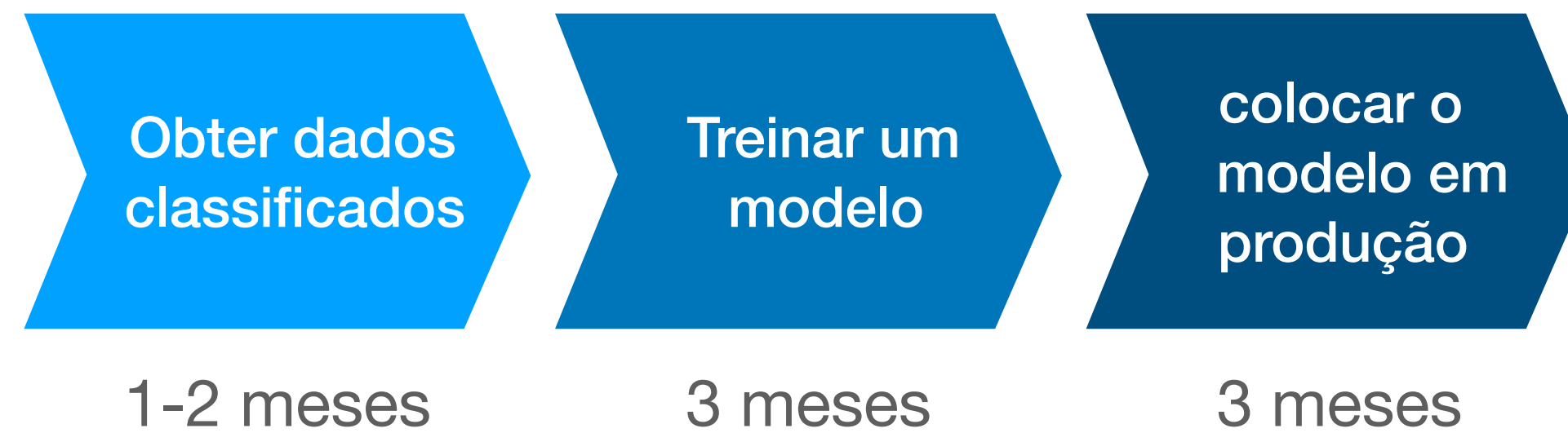
Nota: parte dos slides desta secção são inspirados e adaptados da apresentação de [Andrew Ng: Opportunities in AI - 2023](#), disponível no Youtube.



Prompting

A revolucionar o desenvolvimento de aplicações em Inteligência Artificial

Aprendizagem
supervisionada



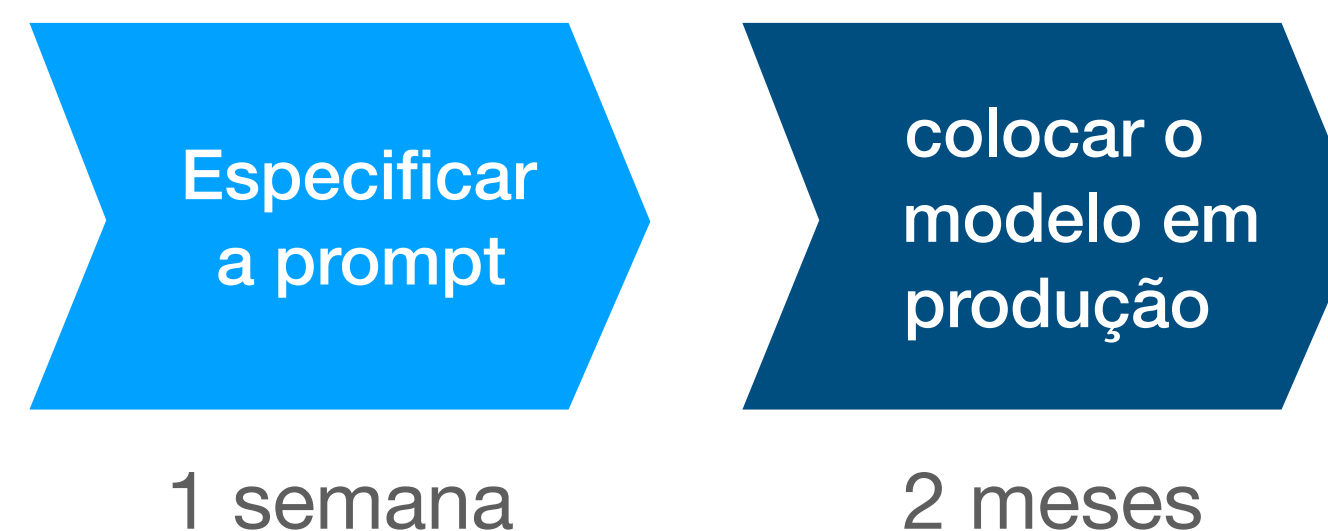
Prompting

A revolucionar o desenvolvimento de aplicações em Inteligência Artificial

Aprendizagem supervisionada



Prompt-based



Exemplo



Classifica o sentimento de cada uma das frases abaixo, iniciadas por asterisco (*), com uma das seguintes possibilidades: positivo, negativo, neutro. Produz uma simples tabela com o resultado, sem explicações adicionais.

- * Gosto muito de estar aqui convosco a falar-vos sobre a Inteligência Artificial, um assunto que me interessa verdadeiramente
- * Se a apresentação se atrasar podemos vir a passar momentos de fome
- * Felizmente, está tudo controlado e vai terminar a tempo
- * Por falar em tempo, será que amanhã chove?

Exemplo



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Frase	Sentimento
Gosto muito de estar aqui convosco a falar-vos sobre a Inteligência Artificial, um assunto que me interessa verdadeiramente	positivo
Se a apresentação se atrasar podemos vir a passar momentos de fome	negativo
Felizmente, está tudo controlado e vai terminar a tempo	positivo
Por falar em tempo, será que amanhã chove?	neutro
Export to Sheets	

Exemplo



Classifica o sentimento de cada uma das frases abaixo, iniciadas por asterisco (*), com uma das seguintes possibilidades: positivo, negativo, neutro. Produz uma simples tabela com o resultado, sem explicações adicionais.

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Frase

Gosto muito de estar aqui convosco a falar-vos sobre a Inteligência Artificial, um assunto que me interessa verdadeiramente

Se a apresentação se atrasar podemos vir a passar momentos de fome

Felizmente, está tudo controlado e vai terminar a tempo

Por falar em tempo, será que amanhã chove?



Aqui está a tabela de classificação de sentimentos para as frases fornecidas:

Frase

Sentimento

Gosto muito de estar aqui convosco a falar-vos sobre a Inteligência Artificial, um assunto que me interessa verdadeiramente

Positivo

Se a apresentação se atrasar podemos vir a passar momentos de fome

Negativo

Felizmente, está tudo controlado e vai terminar a tempo

Positivo

Por falar em tempo, será que amanhã chove?

Neutro

<https://chat.openai.com/> e <https://bard.google.com/>

Exemplo com código

```
In [14]: import openai
import os

from dotenv import load_dotenv, find_dotenv
_ = load_dotenv(find_dotenv()) # read local .env file

openai.api_key = os.getenv('OPENAI_API_KEY')
```

```
In [15]: def get_completion(prompt, model="gpt-3.5-turbo"):
    messages = [{"role": "user", "content": prompt}]
    response = openai.ChatCompletion.create(model=model, messages=messages,
        temperature=0, # this is the degree of randomness of the model's output
    )
    return response.choices[0].message["content"]
```

```
In [27]: prompt = """Classifica o sentimento de cada uma das frases abaixo, iniciadas por asterisco (*),
com uma das seguintes possibilidades: positivo, negativo, neutro.
Produz uma tabela com o resultado no formato csv, sem explicações adicionais.

* Gosto muito de estar aqui convosco a falar-vos sobre a Inteligência Artificial,
um assunto que me interessa verdadeiramente
* Se a apresentação se atrasar podemos vir a passar momentos de fome
* Felizmente, está tudo controlado e vai terminar a tempo
* Por falar em tempo, será que amanhã chove?
"""

response = get_completion(prompt)
print(response)
```


Exemplo com código

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```

```
In [27]: prompt = """Classifica o sentimento de cada uma das frases abaixo, iniciadas por asterisco (*),
com uma das seguintes possibilidades: positivo, negativo, neutro.
Produz uma tabela com o resultado no formato csv, sem explicações adicionais.

* Gosto muito de estar aqui convosco a falar-vos sobre a Inteligência Artificial,
um assunto que me interessa verdadeiramente
* Se a apresentação se atrasar podemos vir a passar momentos de fome
* Felizmente, está tudo controlado e vai terminar a tempo
* Por falar em tempo, será que amanhã chove?
"""

response = get_completion(prompt)
print(response)
```

Frase | Sentimento

Gosto muito de estar aqui convosco a falar-vos sobre a Inteligência Artificial, um assunto que me interessa verdadeiramente | positivo

Se a apresentação se atrasar podemos vir a passar momentos de fome | negativo

Felizmente, está tudo controlado e vai terminar a tempo | positivo

Por falar em tempo, será que amanhã chove? | neutro

Available online services

- Almost all these services require you to register and are paid
- Some available demos
 - <https://monkeylearn.com/sentiment-analysis-online/>
 - some fails: “The unemployment is decreasing”: Negative (74.9%)
 - <https://demo.allennlp.org/sentiment-analysis/roberta-sentiment-analysis>
 - some fails: “prices are increasing”: positive (very confident)
- Your challenge
 - Try finding an online sentiment analysis service that provides a better answer

Summary and conclusions

- Sentiment analysis is a promising topic in the field of marketing and customer relationship management, but also in many other areas.
- Social media and its associated word-of-mouth effect is turning out to be one of the most important sources of information for companies and their customers' sentiments towards their brands and products
- Can be combined with other NLP tasks, such as Topic Classification

Let's check some practical examples (using Python) ...

- Resources



- <https://github.com/HLT-ISCTE/Text-Mining>

Text Mining [↗](#)

Fernando Batista e Ricardo Ribeiro [↗](#)

 [Open in Colab](#)